

NavSafe: Benchmarking Closed-Loop Driving Safety in Photorealistic Environments

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[TODO: FIGURE — TEASER. Left: the evaluation gap—OL benchmarks use real observations but omit sequential feedback; CARLA-based CL benchmarks restore feedback with synthetic imagery; photorealistic CL platforms lack taxonomy-grounded safety coverage. Right: NavSafe combines closed-loop execution, photorealistic observations, and taxonomy-grounded safety-critical events, illustrated with representative capability-level failures.]

Fig. 1. NAVSAFE targets the intersection of closed-loop feedback, photorealistic observations, and taxonomy-grounded safety diagnosis.

Abstract—End-to-end driving policies have progressed rapidly on low-cost open-loop benchmarks, yet driving safety depends on the states a policy induces and the responses it elicits. Open-loop evaluation cannot reveal compounding errors, failure recovery, or interactive behavior. Existing closed-loop benchmarks restore this feedback, but CARLA-based platforms introduce an observation-domain gap, while photorealistic platforms do not organize scenario coverage around an explicit traffic-safety taxonomy. It therefore remains unclear which safety failures current policies exhibit under photorealistic closed-loop execution. We introduce NAVSAFE, a benchmark that combines closed-loop execution, photorealistic observations, and taxonomy-grounded safety-critical events. Guided by the 28 outcome leaves of RoadSafe365, we screen the nuPlan test set for relevant scenarios and supplement selected uncovered taxonomy cells through 3D-asset or dynamic-actor insertion; every admitted scenario is manually verified. NUREC reconstructs each event as an editable environment, and NEXUSSIM executes policies under an event-based, semi-reactive protocol. NAVSAFE reports four event-level metrics and success rates across six safety capabilities. [TODO: Report the numbers of evaluated scenarios and checkpoints, the dominant failure modes, the weakest capabilities, the OL–CL ranking discrepancy, and whether reward-based post-training narrows the gap.] NAVSAFE provides a controlled simulator stress test of closed-loop behavior, not an estimate of real-world crash risk.

I. INTRODUCTION

End-to-end (E2E) driving policies have advanced rapidly on open-loop (OL) benchmarks such as nuScenes and NAVSIM, which provide low-cost, scalable evaluation on real sensor observations [1]–[3]. Driving, however, is sequential: a deployed policy changes its viewpoint, visits off-demonstration states, and influences surrounding traffic. Because an OL prediction does not determine the policy’s next observation or nearby actors’ responses, OL evaluation cannot fully examine accumulated errors, failure recovery, or interaction; policy rankings can consequently change under

closed-loop (CL) execution [4], [5]. Existing CL benchmarks address only part of this evaluation gap. Bench2Drive and Fail2Drive restore interaction in CARLA but use synthetic observations [6], [7], whereas HUGSIM and DriveArena provide photorealistic dynamic traffic without organizing scenario coverage around an explicit real-world safety taxonomy [8], [9]. As illustrated in Fig. 1, an evaluation gap remains at the intersection of sequential feedback, realistic observations, and auditable safety-behavior coverage.

To address this evaluation gap, we introduce NAVSAFE with three defining properties. **1) Closed-loop execution:** NEXUSSIM repeatedly renders policy-induced observations and advances focal traffic in a semi-reactive mode. **2) Photorealistic observations:** NUREC reconstructs selected real driving windows as editable neural environments [10], [11]. **3) Safety-critical events:** RoadSafe365 provides the traffic-safety taxonomy that guides scenario coverage [12]. We screen the nuPlan test set using relevant scenario labels and supplement selected uncovered taxonomy cells through targeted 3D-asset or dynamic-actor insertion [13]. Every admitted scenario is manually verified for taxonomy alignment and physical plausibility, and its provenance remains explicit. NAVSAFE asks which CL safety failures current E2E policies exhibit under photorealistic observations and whether post-training mitigates them.

The evaluation protocol assigns each RoadSafe365 leaf to one primary capability: longitudinal following, roadway containment, right-of-way negotiation, traffic-control compliance, vulnerable-road-user interaction, or incident and directed-traffic adaptation. Each event is scored with Driving Score, Success Rate, Driving Efficiency, and Comfortness, together with success-rate-only capability scores. We compare publicly released checkpoints spanning offline IL/SFT, offline proxy-reward RL fine-tuning, and online closed-loop RL to determine whether source-benchmark gains transfer to photorealistic CL behavior.

Our contributions are three-fold: 1) NavSafe Benchmark 2) Findings

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II. RELATED WORK

A. End-to-End Driving Benchmarks

Open-loop benchmarks. Open-loop (OL) benchmarks prevail because they provide a low-cost, scalable, and effective proxy for driving performance on real sensor observations [1]–[3], [14], [15]. nuScenes evaluates predicted trajectories against a recorded future, while NAVSIM and NAVSIM v2 introduce increasingly behavior-aware planning scores [1]–[3]. However, the policy does not continuously act on newly induced observations. OL evaluation therefore cannot fully reveal compounding errors, recovery from failure, or reactions from surrounding traffic, and its rankings can diverge from closed-loop (CL) deployment [4], [5]. This limitation has motivated recent CL benchmarks.

Closed-loop benchmarks. CL benchmarks execute policies through sequential environment feedback. CARLA-based Bench2Drive and Fail2Drive test interactive driving and controlled generalization shifts, but retain an observation-domain gap for policies trained on real imagery [4], [6], [7]. HUGSIM and DriveArena improve photorealism and support dynamic, safety-relevant traffic [8], [9]; however, their scenario suites are not organized around an explicit traffic-safety taxonomy with fixed behavior-wise coverage. Existing CL benchmarks thus leave an *evaluation gap*: they may expose failures without establishing which safety behaviors were systematically tested. NAVSAFE addresses this gap by jointly providing photorealistic CL execution, taxonomy-guided scenario design, and capability-level safety diagnosis.

B. End-to-End Driving Policies

Offline training. Both conventional E2E policies and recent vision–language–action models commonly rely on offline imitation learning (IL) or supervised fine-tuning (SFT). LTF, DiffusionDrive, and DrivoR provide representative public OL-trained checkpoints [2], [14]–[19]. Because these policies learn only from expert-visited observations, they are not trained to recover from their own errors or react to the consequences of their actions. Strong OL planning scores therefore do not imply safe CL behavior.

Reward-based post-training. Recent methods optimize offline proxy rewards or collect online simulator feedback. ReCogDrive and MTDrive release matched SFT/IL and offline-RLFT checkpoints, AutoVLA releases only its final offline-RLFT model, and MindDrive releases an online-RL policy without its pre-RL counterpart [20]–[27]. Yet source-benchmark gains neither establish transfer to photorealistic CL deployment nor identify which safety behaviors improve. As summarized in Table I, NAVSAFE evaluates these training regimes under one safety-grounded protocol, using matched checkpoints where available to test whether post-training narrows the OL–CL evaluation gap.

III. NAVSAFE BENCHMARK

NAVSAFE connects safety-oriented scenario coverage with photorealistic closed-loop evaluation. This section describes

how scenarios are curated from real driving logs or constructed through targeted interventions (Sec. III-A), converted into editable simulation environments (Sec. III-B), and evaluated through capability-level protocols (Sec. III-D).

A. Scenario Curation

Following the RoadSafe365 taxonomy, we organize the scenario pool around the 28 traffic-safety outcome leaves considered in this study [12], as illustrated in Figure 2. We screen the nuPlan test set using relevant semantic scenario labels to identify naturally occurring candidates [13]. Every scenario is manually reviewed by human against the corresponding RoadSafe365 definition using the sensor observations, map context, and actor trajectories.

[TODO: Update descriptions for uncovered cases.] For each uncovered case in RoadSafe365 taxonomy, we select a compatible normal nuPlan scene and insert a static 3D asset, a dynamic actor, or both to instantiate the target safety situation. The constructed episode e^+ is paired with a reference episode e^0 that preserves the route, background, initialization, and non-focal traffic. Every mined or constructed scenario is manually verified for taxonomy alignment, physical plausibility, and correctness of its trigger and goal.

B. Environment Creation

[TODO: Update the nuRec conversion pipeline after you have selected the scenarios, clarify the input and output, in Figure 3]

C. Scenario Statistics and Coverage

[TODO: Update statistics and coverage, with table or figure]

D. Evaluation Protocol

Simulation setting. We execute each scenario as a bounded closed-loop event in NEXUSSIM. The event contract specifies its trigger, goal region, safety predicates, termination order, time budget, and trace evidence. A short log-replay warm-up initializes the policy history before scoring. We use a *semi-reactive* traffic mode: non-focal traffic follows the recorded log, whereas focal actors respond through IDM or a scenario-specific controller [28]. The policy receives only its declared sensor and navigation inputs; privileged simulator state is reserved for traffic control and scoring.

Event-level metrics. We adapt four metrics from Bench2Drive to event-based evaluation [6]. Let $\mathcal{V}$ denote trials satisfying the frozen reconstruction-validity rules and $N_v = |\mathcal{V}|$. **Driving Score (DS)** combines event completion with multiplicative infraction penalties:

$$DS_i = EC_i \prod_{e \in \mathcal{E}_i} p_e, \quad DS = \frac{100}{N_v} \sum_{i \in \mathcal{V}} DS_i, \quad (1)$$

where $EC_i \in [0, 1]$ and each declared penalty $p_e \in [0, 1]$. For **Success Rate (SR)**, $SR_i = 1$ if event i reaches its goal within budget without a disallowed infraction, and 0 otherwise; $SR = 100 N_v^{-1} \sum_{i \in \mathcal{V}} SR_i$. **Driving Efficiency**

[TODO: FIGURE — 4*7=28 representative scenarios.]

Fig. 2. Scenario Illustration. Dynamic videos could be referred in project page.

[TODO: FIGURE — Conversion and Asset/Actor insertion pipeline.]

Fig. 3. Conversion and Asset/Actor insertion pipeline.

(DE) averages the checkpoint-wise ratio of ego speed to nearby-traffic speed and is not reported when that reference is undefined. **Comfortness** is the fraction of temporal segments satisfying the nuPlan-derived acceleration, jerk, yaw-rate, and yaw-acceleration bounds [6], [13]. All metric thresholds are frozen before test evaluation.

Ability-wise Success Rate (ASR). Before policy evaluation, each event is assigned to one primary capability. Following Bench2Drive’s ability-level reporting [6], we define: **1) Longitudinal Following (LF)**, headway regulation and collision-avoiding acceleration [28]; **2) Roadway Containment (RC)**, drivable-corridor tracking and dynamic stability [29]; **3) Right-of-Way Negotiation (RWN)**, merges, crossings, and unprotected turns [30]; **4) Traffic Control Compliance (TCC)**, signs, signals, and traffic rules [31]; **5) Vulnerable-Road-User Interaction (VRU)**, yielding and clearance around vulnerable road users [32]; and **6) Incident and Directed-Traffic Adaptation (IDTA)**, work zones, obstructions, emergency scenes, and human-directed traffic [12], [33]. Let $\mathcal{I}_k$ be the events assigned to capability k and $\mathcal{V}_k = \mathcal{I}_k \cap \mathcal{V}$. We compute

$$\text{ASR}_k = 100 \frac{\sum_{i \in \mathcal{V}_k} \text{SR}_i}{|\mathcal{V}_k|}, \quad k \in \{\text{LF, RC, RWN, TCC, VRU, IDTA}\} \quad (2)$$

ASR depends only on binary success; progress, efficiency, comfort, and partial penalties cannot increase it. We report the six-dimensional ASR vector as the primary multi-ability result, together with each denominator $|\mathcal{V}_k|$, the macro-average $\overline{\text{ASR}} = \frac{1}{6} \sum_k \text{ASR}_k$, and the worst-capability score $\text{ASR}_{\min} = \min_k \text{ASR}_k$.

IV. EXPERIMENTS

A. Experimental Setup

We evaluate frozen, publicly downloadable checkpoints only after their NEXUSSIM adapters pass observation,

coordinate-frame, horizon, and tracker tests. Offline IL supplies compact trajectory controls; SFT supplies VLA controls. We distinguish *offline proxy-reward RLFT*, which scores actions on fixed logged observations, from *online closed-loop RL*, which updates from policy-induced simulator rollouts. ReCogDrive and MTDrive release matched IL/SFT and RLFT stages [22]–[25]. AutoVLA and MindDrive release final post-trained models without their pre-RL counterparts [20], [21], [26], [27]; they support final-policy comparisons, not estimates of an RL treatment effect.

The test manifest, rubric versions, adapter code, exclusions, and analysis plan are frozen before test rollouts. We record checkpoint hashes, input contracts, parameters, latency, hardware, and licenses. Models with incompatible camera histories, output spaces, or compute budgets remain in separate tracks. SR and the six ASRs are primary safety outcomes; DS, DE, and Comf. describe quality and conservatism. We report seed-cluster 95% intervals, paired seed-level changes for matched checkpoints, and rank association between NAVSIM and NAVSAFE. NAVSIM v1 PDMS and NAVSIM v2 EPDMS are labeled by devkit and split and are never pooled as one scale. PDM-Closed is a privileged solvability reference, not a ranked E2E policy, because it accesses simulator state [34]. **[TODO: Freeze adapters and populate checkpoint, compute, and reproduced NAVSIM metadata.]**

B. RQ1: What Does NAVSAFE Expose Beyond Open-Loop Evaluation?

We first characterize the benchmark rather than rank policies alone. For each policy, we report terminal reasons—ego-attributable collision, roadway departure, rule violation, deadlock, timeout, runtime failure, or reconstruction invalidity—by leaf, capability, and mined/inserted provenance. NAVSIM score versus NAVSAFE SR shows which

TABLE I
PHOTOREALISTIC CLOSED-LOOP RESULTS FOR PUBLIC CHECKPOINTS AND A PRIVILEGED REFERENCE.

Policy	Regime	DS	SR	DE	Comf.	LF	RC	RWN	TCC	VRU	IDTA	Avg.	Min.
<i>Privileged reference</i>													
PDM-Closed*	Oracle	-	-	-	-	-	-	-	-	-	-	-	-
<i>Offline IL/SFT controls</i>													
LTF	IL	-	-	-	-	-	-	-	-	-	-	-	-
DiffusionDrive	IL	-	-	-	-	-	-	-	-	-	-	-	-
DrivoR	IL	-	-	-	-	-	-	-	-	-	-	-	-
ReCogDrive-2B-IL [†]	SFT	-	-	-	-	-	-	-	-	-	-	-	-
MTDrive-SFT [‡]	SFT	-	-	-	-	-	-	-	-	-	-	-	-
<i>Offline proxy-reward RLFT</i>													
ReCogDrive-2B-RL [†]	RLFT	-	-	-	-	-	-	-	-	-	-	-	-
MTDrive-mtGRPO [†]	RLFT	-	-	-	-	-	-	-	-	-	-	-	-
AutoVLA [‡]	RLFT	-	-	-	-	-	-	-	-	-	-	-	-
<i>Online closed-loop RL</i>													
MindDrive-3B [‡]	CL-RL	-	-	-	-	-	-	-	-	-	-	-	-

Avg./Min.: macro-average/minimum ASR. *: privileged reference, not ranked. †: released matched SFT/IL-RLFT pair. ‡: final RL checkpoint only.

failures are absent from one-shot or pseudo-closed-loop scoring. Paired e^0/e^+ failure flips isolate sensitivity to the inserted factor, while representative cases show rendered observations, actor and ego trajectories, the executable predicate, and its trace evidence.

RQ1 also audits the benchmark itself. We report candidate rejection flow, capability-mapping agreement, scorer audit precision/recall, capability correlations, oracle solvability, and reconstruction-invalid rates. Replay-versus-semi-reactive and reconstruction-envelope sensitivity reveal whether policy rankings depend on modeling choices. A new benchmark challenge is supported only when failures recur across independent seeds and the conclusion survives these audits; a uniformly lower aggregate score is insufficient. **[TODO: Insert failure distributions, validity audits, sensitivity results, and representative cases.]**

C. RQ2: Does Post-Training Narrow the OL-CL Gap?

We next test whether gains produced by proxy-reward or online post-training transfer to NAVSAFE. ReCogDrive and MTDDrive provide the primary controlled contrasts: architecture and scale are fixed within each released pair, and per-seed RLFT-minus-SFT changes are reported with 95% intervals. These contrasts estimate the change between released stages, not a universal causal effect of RL, because post-training data and checkpoint selection also change. MindDrive and AutoVLA are final-policy evidence only because their matched pre-RL checkpoints are unavailable.

The analysis never treats VLA and RL as interchangeable. We report exact NAVSIM protocol scores, rank association with NAVSAFE, matched post-training changes, and descriptive capability profiles for unmatched checkpoints. Improvements in SR or ASR support safety transfer within the modeled scenarios; improvements confined to DS, DE, or Comf. indicate better quality or conservatism, not safer

capability. Category means over the few heterogeneous VLA models are not used for inference. **[TODO: Insert matched RLFT deltas, rank association with uncertainty, and the answer to RQ2.]**

Reproducibility. The release will include the taxonomy codebook, candidate and rejection manifests, source tokens, scenario programs, reconstruction quality cards, trace schema, scorer tests, exact checkpoint hashes, adapters, and per-episode results. Development scenarios support adapter debugging; final test manifests remain held out until the evaluation configuration is frozen. Restricted source imagery or model weights are referenced by immutable identifiers and acquisition scripts rather than redistributed. **[TODO: Insert repository, benchmark version, artifact licenses, and evaluation compute.]**

V. CONCLUSION, LIMITATIONS, AND FUTURE WORK

NAVSAFE reframes photorealistic closed-loop evaluation as a coverage, attribution, and validity problem. It converts a real-world safety taxonomy into auditable event contracts, constructs mined and paired-intervention tests from nuPlan, and evaluates public policies in NEXUSSIM with success-rate-only capability profiles, driving-quality metrics, and unscored safety margins. **[TODO: State the final empirical answers to RQ1-RQ2, including effect sizes and uncertainty, and the release scope in two to three sentences.]**

The benchmark provides validity evidence within the declared simulator, not proof of a real-world safety construct. Taxonomy coverage and mapping agreement support content interpretation; real seeds support ecological relevance; paired interventions support attribution; manual scorer audits support measurement reliability; and capability correlations test, but cannot guarantee, discriminant structure. None establishes criterion validity for real crash risk. NAVSAFE oversamples rare and constructed hazards and has no exposure

[TODO: FIGURE — RQ1 RESULTS. Left: failure-mode distribution by policy. Center: LF, RC, RWN, TCC, VRU, and IDTA success-rate heatmap with seed-bootstrap intervals. Right: representative failures with rendered frames, ego/actor traces, terminal reason, and rubric evidence.]

Fig. 4. Failure modes, capability profiles, and validity audits beyond aggregate planning scores.

[TODO: FIGURE — RQ2 OL-CL GAP. Top: reproduced NAVSIM score versus NavSafe SR, labeled by policy and regime. Bottom: paired ReCogDrive and MTDrive SFT-to-RLFT changes for DS, SR, and the six ASRs with bootstrap intervals.]

Fig. 5. NAVSIM-to-NAVSAFE rank transfer and matched SFT-to-RLFT capability changes.

model, so its score is neither a crash-rate estimate nor an on-road safety certification.

The simulation-to-reality gap remains the principal limitation. Photorealistic rendering does not guarantee correct sensor noise, geometry, contact physics, or human behavior. Novel viewpoints can expose reconstruction artifacts, inserted assets can mismatch illumination, and IDM or scripted actors can react unlike people. A policy may therefore fail because of the simulator or pass because the behavior model is too simple. Quality envelopes, paired references, provenance stratification, replay ablations, and sensitivity analyses reduce this ambiguity but cannot remove it. Reconstructed imagery also inherits the privacy and licensing constraints of its source data; releases must preserve nuPlan terms and avoid exposing identifiable content beyond the source dataset.

Future work will expand geographic and environmental diversity, replace hand-scripted interactions with validated behavior distributions, and add public post-trained policies under input- and compute-matched tracks. Frozen releases should remain unchanged; newly mined failures enter versioned future suites. The central next step is external validation: test whether NAVSAFE capability failures predict failures in independent simulators, proving grounds, and supervised real-world interventions better than aggregate open-loop scores.

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